

CodeSight — Status Report

Team HackHive

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 Blocking — no working product without these

 Demo prep — not code

 Optional / post-hackathon

 Bottom line

CodeSight — Status Report

Team HackHive · Tech Eximius 2026 · 2026-08-29

Measured against `CodeSight_Recommended_Features.pdf`, tier by tier.

Legend: **BE** = backend (built + tested + deployed) · **FE** = frontend UI · **Wired** = the frontend actually calls the backend.

Deployment: Backend live on Railway (`codesight-code-review-production.up.railway.app`), Postgres, 61 tests passing (+10 network tests skipped by default). 1002 exercises (19 curated + 983 generated); generated pool under human review. Frontend on Vercel, running on its own mock data — **not yet connected to the backend**.

Tier 1 — Core guidance loop

#	Feature	BE	FE	Wired	Notes
1	Real-code review surface	done — <code>/exercises</code> , <code>/exercises/{id}</code>	done — Monaco workspace	no	FE reads <code>frontend/src/data/exercises.ts</code>
2	Teaching grader (where / why-missed / pattern)	done — <code>/grade</code> (Gemini + fallback)	done — results page	no	
3	Named six-class taxonomy	done — 1002 exercises tagged	done — <code>defectClasses.ts</code>	no	injection, auth, error-handling, concurrency, logic, resource
4	Learn the Concept	done — <code>/concepts</code> , <code>/concept/{id}</code>	done — concept page	no	6 concepts: summary, bad/good, YouTube links (all verified 2026-08-29), practice ids
5	Prescriptive weakness profile	done — <code>/profile</code>	done — dashboard	no	"weakest area X, catch rate Y%, 3 more queued"

Tier 1: backend 5/5, frontend UI 5/5, wired 0/5.

Tier 2 — High value, low cost

#	Feature	BE	FE	Wired	Notes
6	Progressive hints + score decay	done — <code>/exercises/{id}/hints/{n}</code> , <code>hints_used</code> in <code>/grade</code> (1.0 / 0.9 / 0.75 / 0.5)	done	no	
7	Concept micro-check (2-3 questions)	done — <code>GET/POST</code> <code>/concept/{id}/micro-check</code> (3 MCQs/concept, stateless grading, pass = 2/3)	UI exists	no	answer key stays server-side; result returns per-question explanations
8	Progress-over-time	done — <code>/progress/{session_id}</code> (timeline + running catch-rate + per-class first-vs-latest)	done — Recharts	no	
9	Prompt-injection guard	done — grader system prompt + 4000-char cap	n/a	yes (server-side)	basic form

Tier 2: backend 4/4, frontend 3/4, wired 0/4.

Tier 3 — Demo & engagement

#	Feature	BE	FE	Wired	Notes
10	Multiplayer Code Review Battle	not built (plan cut it)	simulated (5 fake bots)	n/a	real version = websocket rooms, ~5-8 person-days
11	Leaderboard	not built	simulated	n/a	depends on 10
12	AI Reviewer vs You	done — <code>/ai-review</code> (blind review, diff, headline, retry)	done — page	no	

Tier 3: backend 1/3, frontend 3/3 (2 simulated), wired 0/3.

Tier 4 — Later, if time allows

#	Feature	BE	FE	Wired	Notes
13	False-positive trap exercises	done — 176 clean exercises + FP penalty in scoring	done	no	flagging clean code -> <code>false_positive</code> , score 0
14	Large-code topic prediction	not built	not built	no	separate exercise type + scoring
15	Exercise feedback loop	done — <code>/exercises/{id}/report</code> (3 distinct sessions -> hidden)	no report button in FE	no	

Tier 4: backend 2/3, frontend 1/3, wired 0/3.

Integrity / anti-cheat section

Layer	Item	Status
1 — Editor friction	disable copy / cut / context-menu	FE has paste notification + char limit + timer per its README; full lockdown unclear
2 — Behavioural detection	paste detection, keystroke cadence, tab-blur, time-to-submit, known-AI match, cross-student similarity, integrity score	none on backend. <code>/grade</code> accepts no telemetry
3 — Task design	per-exercise timer, localise-first, word cap, follow-up probe	word cap done; timer / localise-first are FE; follow-up probe not built
4 — Positioning	"practice not exam" framing	copy decision, not code

Integrity: ~15% done — the biggest untouched section from the doc.

”Getting to 10/10” section

Item	Status
Taxonomy named + documented	done
Determinism (temp 0, hash cache, fixed rubric)	done
Grader benchmark (gold set + agreement number)	not run
Clean-file sourcing from refactor commits	hand-written / generated instead
Confirmed vs plausible findings	in grader prompt, not a structured field
Corpus to 120+	done — 1002 (19 curated + 983 generated; generated pool unreviewed, teammate reviewing via <code>review_exercises.py</code>)
Predictive weakness profile	done; shareable “review-skill card” export not built
Real usage data (test on 8-10 students)	not done
90-second demo loop, rehearsed	not done
Pre-answer judge questions slide	not done

Beyond the doc — the tier system (added later)

Feature	Status
beginner / intermediate / pro tiers	done, deployed
Cumulative content gating (<code>/exercises?tier=</code>)	done
Session tier (<code>/session/{id}</code> , auto-created at beginner)	done
Promotion test (3 curated next-tier exercises)	done — <code>/promotion-test/{id}</code>
Promotion evaluate (first-attempt mean ≥ 0.7 -> promote)	done — verified live
Frontend UI for tiers / promotion	not built

What's left to finish the website

Blocking — no working product without these

#	Task	Owner	Size
A	Wire the frontend to the backend. Replace <code>frontend/src/data/*.ts</code> with real calls to <code>/exercises</code> , <code>/grade</code> , <code>/profile</code> , <code>/progress</code> , <code>/concept/{id}</code> , <code>/session/{id}</code> , <code>/promotion-test/*</code> , <code>/exercises/{id}/hints/{n}</code> . Generate <code>sessionId</code> -> <code>localStorage</code> .	FE pair	~1 day
B	Tier UI: badge, tier-gated practice list, “take promotion test” flow	FE pair	~half day
C	“Report exercise” button (calls <code>/exercises/{id}/report</code>) — needed because 983 generated exercises are unvalidated (human review in progress via <code>review_exercises.py</code> + <code>exercises.review.json</code> sidecar)	FE pair	~1 hr
D	End-to-end test on the deployed stack (Vercel -> Railway), fix breakage	Both	~3 hrs
E	Verify the 6 concept YouTube URLs are live DONE (2026-08-29) — all 10 links checked via YouTube oEmbed; 2 dead links replaced, 4 titles corrected (commit <code>b22aced</code>)	Anyone	—

Demo prep — not code

#	Task
F	Decide + rehearse the 90-second demo journey (review -> weak concept -> video -> practice -> promote)
G	Pitch deck + roadmap slide (deferred: code-writing track, multiplayer, integrity telemetry)
H	Judge-question answers (grader accuracy, how it scales, why 6 classes, gaming)

Optional / post-hackathon

#	Task	Note
I	Wire the micro-check UI to <code>GET/POST /concept/{id}/micro-check</code> (backend done)	FE, small
J	Integrity telemetry on <code>/grade</code> (paste events, timing, integrity score)	the whole Integrity section
K	Multiplayer + leaderboard (#10, #11)	5-8 person-days — roadmap only
L	Human-review the 983 generated exercises (<code>review_exercises.py</code> , beginner tier first)	teammate, ongoing
M	Grader benchmark (gold set + agreement %)	strong for judges
N	Large-code topic prediction (#14)	roadmap

Bottom line

Backend: ~92% of the recommended feature set is built, tested, and live — everything except multiplayer, large-code prediction, and integrity telemetry (all explicitly deferrable).

Frontend: UI exists for most features but runs entirely on mock data.

The single thing standing between “impressive backend” and “working product” is task A — connecting the two. Everything else is polish or roadmap.